

Rings are understood to be commutative, unless stated otherwise.

Problem 1

Let R be an algebra over a field k . If R is a domain and R has finite dimension as a k -vector space, then R is a field.

Proof: $\forall a \in R/\{0\}$, let $\varphi_a : R \rightarrow R$ where $x \rightarrow ax$, note that $\varphi_a(R) = aR$. Additionally, note the following for all $\alpha, \beta \in k$ we have that $\varphi_a(\alpha x + \beta y) = a(\alpha x + \beta y) = \alpha ax + \beta ay = \alpha \varphi_a(x) + \beta \varphi_a(y)$. Hence $\varphi \in \text{End}_k(R)$. $\forall k \in \ker \varphi_a$, $0 = \varphi_a(k) = ak$. Since R is a domain and $a \neq 0$, $k = 0$. Thus $\ker \varphi_a = \{0\}$ and φ_a is injective. Since R is finite dimensional as a k -vector space, any injective linear endomorphism must also be surjective. Thus $R \subset \varphi_a(R) = aR$. Now consider any nonzero ideal I , $\exists a \in I$ nonzero, thus $R \subseteq aR \subseteq I \subseteq R \Rightarrow I = R$. Thus the only ideals of R , are 0 and R . Hence R is a field. $\square$